

NAME -: VAIBHAV SINGH
BRANCH -: AI-DS
ROLL NO -: 2301321630047

1 Task

```
Code Coverage: None API Version: 67
1 trigger NoDuplicateEmail on Contact (before insert) {
2   if (Trigger.isBefore && Trigger.isInsert) {
3
4     Set<String> emailSet = new Set<String>();
5     for (Contact con : Trigger.new) {
6       if (con.Email != null) {
7         emailSet.add(con.Email);
8       }
9     }
10
11
12     Map<String, Contact> existingContactsMap = new Map<String, Contact>();
13     if (!emailSet.isEmpty()) {
14       for (Contact c : [SELECT Id, Email FROM Contact WHERE Email IN :emailSet]) {
15         existingContactsMap.put(c.Email, c);
16       }
17     }
18
19     for (Contact con : Trigger.new) {
20       if (con.Email != null && existingContactsMap.containsKey(con.Email)) {
21         con.Email.addError('A Contact with this email already exists.');
```

* = Required Information

Contact Information

Contact Owner: Vaibhav Singh

Phone: 09129019491

Home Phone:

* Name:

Salutation: Mr.

First Name: vaibhav

* Last Name: singh

Account Name: Search Accounts...

Mobile:

Title:

Other Phone:

Department:

Fax:

Birthdate:

Email: singhvaibhav849@gmail.com
A Contact with this email already exists.

Reports To: Search Contacts...

Assistant:

Lead Source: --None--

Profile:

⚠ We hit a snag.

Review the following fields

- Email

Cancel Save & New Save

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2 Task

The screenshot shows the Salesforce Developer Console with an Apex trigger named 'AutoPopulate' on the 'Contact' object. The trigger is configured to run 'before insert, after insert, after delete'. The code logic is as follows:

```
1 trigger AutoPopulate on Contact (before insert, after insert, after delete) {  
2     if (Trigger.isBefore && Trigger.isInsert) {  
3         Set<String> emailSet = new Set<String>();  
4  
5         for (Contact con : Trigger.new) {  
6  
7             if (con.Email != null) {  
8                 emailSet.add(con.Email);  
9             }  
10            if (con.Title == 'Manager') {  
11                con.Department = 'Management';  
12            } else if (con.Title == 'Developer') {  
13                con.Department = 'IT';  
14            } else if (con.Title == 'Tester') {  
15                con.Department = 'QA';  
16            } else if (con.Title == 'HR') {  
17                con.Department = 'Human Resources';  
18            } else {  
19                con.Department = 'General';  
20            }  
21        }  
22    }  
23 }
```

On the right sidebar, a preview of a Contact record is shown:

- Contact Owner: Vaibhav Singh
- Name: Mrs. xyq abc
- Account Name: 25896314
- Title:
- Department:

The screenshot shows the Salesforce record page for 'Mr. akhay singh'. The page includes a 'Details' tab and an 'Activity' section.

Contact Details:

Field	Value
Contact Owner	Vaibhav Singh
Name	Mr. akhay singh
Account Name	rudra
Title	
Department	General
Birthdate	
Reports To	
Lead Source	
Profile	
Revenue	
Status	
New_Field	
Mailing Address	
Languages	
Phone	
Home Phone	
Mobile	
Other Phone	
Fax	
Email	
Assistant	
Asst. Phone	
Other Address	
Level	

Activity Section:

- Filters: All time • All activities • All types
- Buttons: Refresh, Expand All, View All
- Section: Upcoming & Overdue
- Status: No activities to show.
- Message: Get started by sending an email, scheduling a task, and more.
- Tip: To change what's shown, try changing your filters.
- Button: Show All Activities

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Task 3

Student_Management

Home

Account

Accounts

✓ Contact "praksh singh" was created.

Courses

Job Applications

More

Account

rudra

+ Follow

New Contact

New Case

New Note

Account Number 14785236	Website
Account Site	Ticker Symbol
Type	Ownership
Industry	Employees
Education	SIC Code
Annual Revenue \$65,000	
Vaibhav Singh	
Active Account ✓	
Profile	
Total_Contacts 2	
Billing Address Greter noida Greter noida greter noida, Uttar Pradesh 201310 India	Shipping Address Greter noida Greter noida greter noida, 201310
Customer Priority	SLA
SLA Expiration Date	SLA Serial Number
Number of Locations	Upsell Opportunity
Active	

Employee__c@10:06 AM Employee__c@10:06 AM Employee__c@10:06 AM Employee__c@10:06 AM Employee__c@10:06 AM TrgContUpdate.apxt ContactTrigger.apxt ContactsTrigger.apxt * Contacts.apxt contacts.apxc

Code Coverage: None API Version: 67

```
1 trigger ContactCreated on Contact (after insert, after delete) {
2     Set<Id> accountIds = new Set<Id>();
3     List<Contact> contactsList = (Trigger.isInsert) ? Trigger.new : Trigger.old;
4     for (Contact con : contactsList) {
5         if (con.AccountId != null) {
6             accountIds.add(con.AccountId);
7         }
8     }
9 }
10 if (!accountIds.isEmpty()) {
11
12     Map<Id, Account> accountsMap = new Map<Id, Account>([
13         SELECT Id, Total_Contacts__c, (SELECT Id FROM Contacts)
14         FROM Account
15         WHERE Id IN :accountIds
16     ]);
17     // Case 1 -: whenever new contact is created then child have count -: if we print child count we get the count data
18     // Case 2 -: if delete contact delete count the child give all present count , access this count
19
20     List<Account> accountsToUpdate = new List<Account>();
21     for (Id accId : accountIds) {
22         Account acc = accountsMap.get(accId);
23         if (acc != null) {
24             List<Contact> childContacts = acc.Contacts;
25             acc.Total_Contacts__c = (childContacts != null) ? childContacts.size() : 0;
26             accountsToUpdate.add(acc);
27         }
28     }
29     if (!accountsToUpdate.isEmpty()) { //Update DML
30         update accountsToUpdate;
31     }
32 }
```

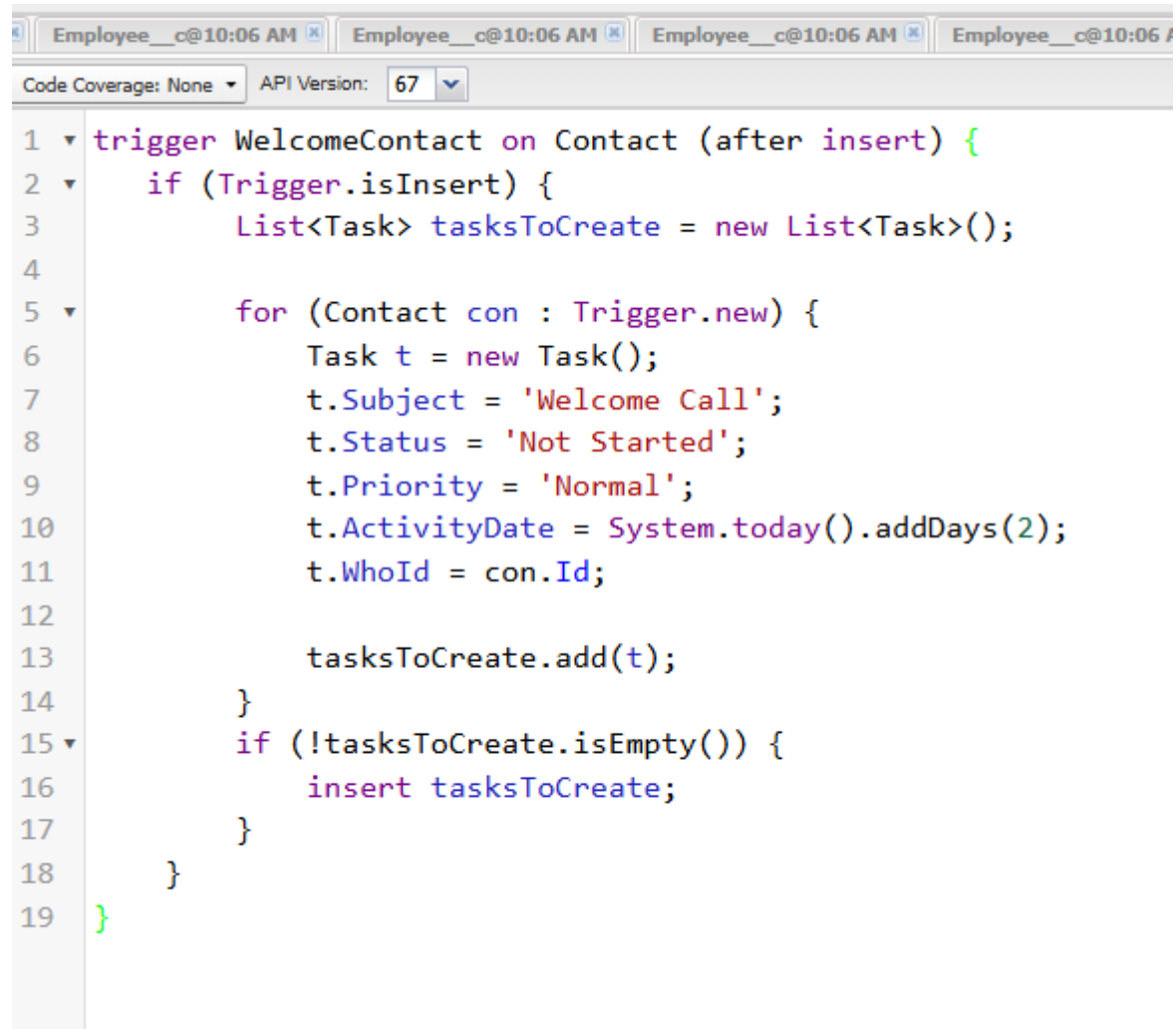
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Task 4

The screenshot displays the 'Student Management' interface. The top navigation bar includes links for Home, Account, Accounts, Contacts, Reports, Dashboards, Students, Courses, Job Applications, Departments, Employees, Leave Requests, and E-Comm Orders. The main header shows the contact name 'Ms. Suman Singh' with a profile icon and a 'Follow' button. Below the header, there are tabs for 'Related' and 'Details'. The 'Related' tab is active, showing a message: 'We found no potential duplicates of this Contact.' Below this message are four sections: 'Opportunities (0)' with a 'New' button, 'Cases (0)' with a 'New' button, 'Campaign History (0)' with an 'Add to Campaign' button, and 'Notes & Attachments (0)' with an 'Upload Files' button. The 'Activity' tab is also visible, showing a list of activities under the heading 'Upcoming & Overdue'. The first activity is 'Welcome Call' with a status of 'Upcoming' and a date of 'Aug 2'. A 'Show All Activities' button is at the bottom of the activity list.

The screenshot shows the same 'Student Management' interface as above, but with a 'Choose Status' modal dialog open. The dialog has a title bar with a close button (X). Inside the dialog, there is a section titled 'Select a Status' with four radio button options: 'Not Started', 'In Progress', 'Waiting on someone else', and 'Deferred'. At the bottom of the dialog are two buttons: 'Cancel' and 'Save'. The background of the interface is dimmed, showing the 'Activity' tab and the 'Upcoming & Overdue' section with the 'Welcome Call' activity.

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```
1  trigger WelcomeContact on Contact (after insert) {  
2      if (Trigger.isInsert) {  
3          List<Task> tasksToCreate = new List<Task>();  
4  
5          for (Contact con : Trigger.new) {  
6              Task t = new Task();  
7              t.Subject = 'Welcome Call';  
8              t.Status = 'Not Started';  
9              t.Priority = 'Normal';  
10             t.ActivityDate = System.today().addDays(2);  
11             t.WhoId = con.Id;  
12  
13             tasksToCreate.add(t);  
14         }  
15         if (!tasksToCreate.isEmpty()) {  
16             insert tasksToCreate;  
17         }  
18     }  
19 }
```

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Task 5 -:



Success:
Student successfully registered.

Course Registration Portal

▼ Enter Student Information

Student Name

Email

Phone

▼ Available Courses (Select One)

Select	Course Name	Course Fee	Duration
☐	B.tech	\$144,400	4
☐	BBA	\$1,440	4

Register

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Course Registration Portal

▼ Enter Student Information

Student Name

Email

Phone

▼ Available Courses (Select One)

Select	Course Name	Course Fee	Duration
☒	B.tech	\$144,400	4
☐	BBA	\$1,440	4

```
1 <apex:page controller="StudentControllers">
2 <apex:form>
3   <apex:pageMessages/>
4   <apex:pageBlock title="Course Registration Portal">
5     <apex:pageBlockSection title="Enter Student Information" columns="1">
6       <apex:inputField value="{!student.Name}" required="true"/>
7       <apex:inputField value="{!student.Email__c}" required="true"/>
8       <apex:inputField value="{!student.Phone__c}" />
9     </apex:pageBlockSection>
10    <apex:pageBlockSection title="Available Courses (Select One)" columns="1">
11      <apex:pageBlockTable value="{!courseList}" var="c">
12        <apex:column headerValue="Select">
13          <apex:selectRadio value="{!selectedCourseId}">
14            <apex:selectOption itemValue="{!c.Id}" itemLabel="" />
15          </apex:selectRadio>
16        </apex:column>
17        <apex:column value="{!c.Name}" />
18        <apex:column value="{!c.Course_Fee__c}" />
19        <apex:column value="{!c.Duration__c}" />
20      </apex:pageBlockTable>
21    </apex:pageBlockSection>
22    <apex:pageBlockSection columns="1">
23      <apex:commandButton value="Register" action="{!registerStudent}" />
24    </apex:pageBlockSection>
25  </apex:pageBlock>
26 </apex:form>
27 </apex:page>
```

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```
Code Coverage: None API Version: 67
1 public class StudentControllers {
2
3     public List<Course__c> courseList {get;set;}
4     public Student__c student {get;set;}
5     public Id selectedCourseId {get;set;}
6
7     // Constructor
8     public StudentControllers(){
9         student = new Student__c();
10        courseList = [
11            SELECT Id, Name, Course_Fee__c, Duration__c
12            FROM Course__c
13        ];
14    }
15    public PageReference registerStudent(){
16
17        if(selectedCourseId == null){
18            ApexPages.addMessage(new ApexPages.Message(ApexPages.Severity.ERROR, 'Please select a course.));
19            return null;
20        }
21        student.Course__c = selectedCourseId;
22        insert student;
23        ApexPages.addMessage(new ApexPages.Message(ApexPages.Severity.CONFIRM, 'Student successfully registered.));
24
25        student = new Student__c();
26        selectedCourseId = null;
27
28        return null;
29    }
30 }
```